

A Simple Model of Asset Pricing with Rationally Inattentive Investors in a Stochastic Discount Factor Framework

MA Thesis

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Abstract

I revisit a two-period, two-state model of Chavchanidze (2021), in which I analyze the portfolio selection problem of a rationally inattentive investor with uniform priors and fixed prices. In this paper, I relax the two restrictions and describe optimal information strategy for arbitrary prior beliefs, obtain unconditional signal probabilities and posterior beliefs in closed form and discuss asset pricing implications in a stochastic discount factor framework. The results mirror findings in Chavchanidze (2021): even with arbitrary priors and flexible prices, incomplete information induces risk-aversion in the good state and risk-seeking in the bad state relative to the complete information case. I also comment briefly on how this framework can yield insights about risk premium and low risk-free rate puzzles.

Keywords: Rational Inattention, Cost of Information, Risk Premium Puzzle

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1 Introduction

Standard Neoclassical models of asset pricing rely on ultra-rational investors who hold correct beliefs and face no informational frictions. Such models, therefore, conflict with data and generate some well-known empirical puzzles, such as the risk premium puzzle or the low risk-free rate puzzle. Two strands of literature have emerged in response to the shortcomings of the Neoclassical models. Behavioral approach modifies the standard framework by modeling investors as being biased and having non-standard preferences (see Shefrin (2008) for a thorough review of this approach), while the bounded rationality approach does not a priori model investors as suboptimal, but assume that they face information processing costs, which induce behavior different from what is implied by the standard Neoclassical models (see e.g. Batchuluun et al. (2019), Luo (2006) and Mackowiak et al. (2020)). In the introductory section of Chavchanidze (2021), I demonstrate how these approaches are different yet complementary and build a case for their merger. More specifically, I propose incorporating Rational Inattention (RI) approach - a model of bounded rationality - into a behavioral stochastic discount factor framework. I provide a mathematical motivation and justification for the merger and derive some of its basic properties. I also discuss a portfolio selection problem in the case of log-utility, 2 states, uniform priors and fixed prices (Section 4), and illustrate how the implications of this framework differ from those of the standard neoclassical approach.

In this paper, I extend the analysis in Section 4 of Chavchanidze (2021) by relaxing the assumptions of fixed prices and uniform priors, and analyze the asset pricing implications of costly information. The contribution of this paper is three-fold. Firstly, it illustrates some of the basic mechanisms of the model proposed in Chavchanidze (2021) in a simple setting and in doing so, lays the foundation for understanding more complex formulations that do not rely on two states or two periods. Secondly, it demonstrates how the rational inattention approach can be incorporated into the stochastic discount factor framework of asset pricing, a feat not yet attempted to the best of my knowledge (even though the model was originally formulated in Chavchanidze (2021) in a more general setting, asset pricing implications have not been discussed before). Thirdly, I obtain unconditional signal probabilities and posterior beliefs in closed form and for general 2-state distribution. This could greatly improve the tractability of the 2-state 2-period model, which could be very useful in itself. For instance, it can help provide insights about how the cost of information might affect the risk free rate or the risk

premium, or how the incompleteness of information induced by RI might interact with pre-existing behavioral biases, such as inaccuracy of prior beliefs about tomorrow or the representativeness heuristic (even though this paper does not deal with behavioral elements, a reader might refer to Sections 4.4. and 5.2 in Chavchanidze (2021) to see how this framework can be used to analyze such biases in the context of imperfect information).

2 Model Description

Consider a continuum of identical investors who want to allocate their wealth across consumption today and consumption tomorrow. For each $t \in \{0, 1\}$, there are two possible states at time t - a good state (x_t^2) and a bad state (x_t^1), $t = 0, 1$. In the good time-zero state x_0^2 , the probability of the good state x_1^2 tomorrow is $p > \frac{1}{2}$, while in the bad time-zero state x_0^1 , the probability of the good state x_1^2 tomorrow is $1 - p$. The prior probability of the good time-zero state x_0^2 is $p_0 > \frac{1}{2}$. There are two assets the representative investor can use to facilitate consumption in the two states tomorrow - a risky asset and a safe asset. The risky asset pays X_j units of consumption in state x_2^j tomorrow, $j = 1, 2$, $X_2 = 1 + a$, $X_1 = 1 - a$, $0 < a < 1$, while the safe asset pays one unit of consumption in both states tomorrow. The economy is endowed with ω_0 units of consumption today and $\omega(x_1^j)$ units of consumption in state x_1^j tomorrow, $j = 1, 2$, $\omega(x_1^2) > \omega(x_1^1)$.

The representative investor has time-separable log-preferences over consumption and wants to allocate her wealth across time-zero consumption and the two assets in a way that maximizes her two-period expected utility. However, the probability distribution over the two states tomorrow depends on the current state, which is unobservable. The agent, therefore, maximizes her expected utility with respect to posterior beliefs induced by a binary noisy signal s_λ that informs about the current state x_0 . A realization s_2 of s_λ suggests that the current state is good (x_0^2), while a realization s_1 suggests that the current state is bad (x_0^1). $\lambda \geq 0$ is a cost of information parameter that indicates the costs of learning in terms of incurred disutility. Higher values of λ correspond to higher costs and $\lambda = 0$ corresponds to the complete information case in which x_0 is fully observable.

The agent chooses s_λ in the Rational Inattention (RI) framework of Matejka and McKay (2015) (see Sections 2.4 of Chavchanidze (2021) for mathematical details). For

now it suffices to say that s_λ maximizes the ex-ante usefulness of the information strategy net of the cost of information, which is a linear function of λ . The unconditional probability of observing s_2 under optimal information strategy is $p_{s,\lambda}$. If $p_{s,\lambda} \in \{0,1\}$, signals are completely uninformative, in which case the representative investor does not process any information and optimizes consumption using her prior beliefs.

A distinctive feature of the RI approach is that the form of signals to choose from is not exogenously specified - the agent optimizes information strategy by choosing among joint distributions of a binary signal s and time-zero state x_0 and has thus an implicit access to any information strategy that she deems feasible.

3 Asset Pricing Implications

3.1 Asset Pricing with Arbitrary Beliefs

Consider the following choice problem: A representative investor wants to allocate her wealth W across time-zero consumption and the safe and risky assets in a way that maximizes her two-period expected utility of consumption, where the expectation is taken w.r.t. beliefs $(1 - \tilde{p}, \tilde{p})$ about (x_1^1, x_1^2) . The optimization problem is as follows:

$$\max_{C_0, a_r, a_f} \{u(C_0) + \beta E_{\tilde{p}}[u(C_1)]\}$$

subject to

$$W - C_0 = a_r P_r + a_f P_f$$

$$C_1 = a_r X + a_f,$$

where a_r and a_f are, respectively, the amount of the risky and safe assets purchased and X is a stochastic pay-off of the risky asset. The subscript $\tilde{p}$ on $E_{\tilde{p}}[\cdot]$ indicates the dependence of the expectation operator on the probability $\tilde{p}$ of the good state occurring. The first-order conditions of the optimization problem w.r.t a_r and a_f give, respectively,

$$-u'(C_0)P_r + \beta E_{\tilde{p}}[u'(C_1)X] = 0$$

$$-u'(C_0)P_f + \beta E_{\tilde{p}}[u'(C_1)] = 0,$$

which imply

$$P_r = \beta E_{\tilde{p}} \left[\frac{u'(C_1)}{u'(C_0)} X \right]$$

$$P_f = \beta E_{\tilde{p}} \left[\frac{u'(C_1)}{u'(C_0)} \right].$$

In equilibrium, $C_0 = \omega_0$ and $C_1^j = \omega(x_1^j)$, $j = 1, 2$ and so the equilibrium prices are given by

$$P_r = \beta E_{\tilde{p}} \left[\frac{u'(\omega(x_1))}{u'(\omega_0)} X \right]$$

$$P_f = \beta E_{\tilde{p}} \left[\frac{u'(\omega(x_1))}{u'(\omega_0)} \right].$$

With $u(c) = \log c$,

$$\frac{u'(\omega(x_1))}{u'(\omega_0)} = \frac{\omega_0}{\omega_1} \equiv \frac{1}{g},$$

where g is a gross aggregate consumption growth rate. Denote

$$\frac{\omega_0}{\omega(x_1^j)} \equiv \frac{1}{g_j}, \quad j = 1, 2$$

and note that in equilibrium

$$P_f = \beta E_{\tilde{p}} \left[\frac{u'(\omega(x_1))}{u'(\omega_0)} \right] = \beta \left((1 - \tilde{p}) \frac{1}{g_1} + \tilde{p} \frac{1}{g_2} \right) \quad (3.1)$$

$$P_r = \beta E_{\tilde{p}} \left[\frac{u'(\omega(x_1))}{u'(\omega_0)} X \right] = \beta \left((1 - \tilde{p}) \frac{1}{g_1} X_1 + \tilde{p} \frac{1}{g_2} X_2 \right). \quad (3.2)$$

3.1.1 Responses of Equilibrium Prices to Changes in Uncertainty

If we measure the degree of uncertainty inherent in the economy by the variance of the probability distribution over the states of the world tomorrow, then the level of uncertainty is described by $\tilde{p}(1 - \tilde{p})$. Conditional on $\tilde{p} > \frac{1}{2}$, the uncertainty decreases as the probability of the good state increases and vice versa. Consider now the partial derivatives of the equilibrium prices of the two assets w.r.t $\tilde{p}$:

$$\frac{\partial P_f}{\partial \tilde{p}} = \beta \left(\frac{1}{g_2} - \frac{1}{g_1} \right)$$

$$\frac{\partial P_r}{\partial \tilde{p}} = \beta \left(\frac{1}{g_2} X_2 - \frac{1}{g_1} X_1 \right).$$

Note that $\frac{\partial P_f}{\partial \tilde{p}} < 0$ unequivocally, i.e. when the probability of the good state increases (equivalently, when the degree of uncertainty falls), the equilibrium value of the safe asset decreases. The response of P_r to an increase in $\tilde{p}$ depends on the sign of $\frac{1}{g_2}X_2 - \frac{1}{g_1}X_1$, where $X_2 > X_1$ and $g_2 > g_1$. $\frac{1}{g_2}X_2$ is the marginal utility of owning an additional ϵ -amount of the risky asset in the good state tomorrow, while $\frac{1}{g_1}X_1$ is the marginal utility of owning an additional ϵ -amount of the risky asset in the bad state tomorrow. Whenever $\frac{1}{g_2}X_2 < \frac{1}{g_1}X_1$, the agent has enough resources for consumption in state x_1^2 and does not want to ensure more consumption in the good state tomorrow relative to the consumption in the bad state tomorrow despite the increasing likelihood of the good state. In that case, increase in $\tilde{p}$ motivates the agent to increase consumption today relative to planned consumptions in the two states tomorrow, which causes the values of both assets to fall. Conversely, whenever $\frac{1}{g_2}X_2 > \frac{1}{g_1}X_1$ the agent values additional consumption in the good state tomorrow relative to the consumption in the bad state and so increase in $\tilde{p}$ encourages risk-taking, which in turn causes the price of the risky asset to increase.

The wedge between P_r and P_f , however, widens unambiguously in response to an increase in $\tilde{p}$:

$$\frac{\partial (P_r - P_f)}{\partial \tilde{p}} = \beta \left(-\frac{1}{g_1}(X_1 - 1) + \frac{1}{g_2}(X_2 - 1) \right) > 0, \quad (3.3)$$

where the inequality follows by $X_2 - 1 = -(X_1 - 1)$. (3.3) can be interpreted as either a consequence of an increased likelihood of the good state or as a consequence of decreased volatility.

3.2 Optimal Signals and Posterior Beliefs

The optimal information strategy is described by the following Proposition:

Proposition 3.1. *Let $p_{s,\lambda}$ denote the probability of observing a good signal under the optimal information strategy and the cost of information λ and let $p_0 > \frac{1}{2}$ be the prior probability of the good state at time 0. Then,*

- i) $\exists \bar{\lambda} > 0$ such that the agent does not process any information for $\lambda \geq \bar{\lambda}$.*
- ii) $\lim_{\lambda \rightarrow 0} p_{s,\lambda} = p_0$,*

$$\text{iii) For small enough } \lambda, p_{s,\lambda} = \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL} \right) - e^{-\frac{\beta}{\lambda}KL}}{1 - e^{-\frac{\beta}{\lambda}KL}},$$

iv) For small enough λ , $\frac{\partial p_{s,\lambda}}{\partial \lambda} > 0$.

Proof. See Part A of the Appendix. $\square$

i) indicates that for high values of λ , the agent optimizes consumption using her prior beliefs. ii) reflects a more general result in Chavchanidze (2021) (Lemma 3.4), according to which signals are perfectly informative in the limit. ii) and iv) together imply $p_{s,\lambda} > p_0 \forall \lambda > 0$ and that signal informativeness falls with increasing costs. In words, the information strategy can be described as follows: when information is very costly to acquire ($\lambda > \bar{\lambda}$), the agent chooses not to process any information. When λ is small enough, she selects signals that are biased towards the good time-zero state and the bias widens as the cost of information increases, i.e. the agent becomes less competent at identifying bad time-zero states.

The optimal information strategy induces posterior beliefs that are described in the Proposition that follows:

Proposition 3.2. *Let s_j denote a signal realization that suggests the current state is j , $j = 1, 2$. Then, for small enough λ*

$$P_{x_1|s_\lambda}(x_1^2|s_2) = (1-p) \frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}} + p \frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}}$$

and

$$P_{x_1|s_\lambda}(x_1^2|s_1) = (1-p) \frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} + p \frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}}.$$

Proof. See Part B of the Appendix. $\square$

Proposition 3.2 says that the posterior probability of the good state tomorrow is a convex combination of $1-p$ and p , the conditional probabilities of the good state induced by different time-zero states. When a good signal realization is observed, the probability induced by the good time-zero state receives a higher weight and vice versa. Weights in turn depend on λ , with smaller values inducing sharper reweighting in response to signal realizations.

In Section 4.2 of Chavchanidze (2021), I obtain the same posterior beliefs in the case of uniform priors. The proof of this proposition holds for arbitrary $p_0 > \frac{1}{2}$. This is a striking result, since it implies that even if the agent has biased priors, the optimal information strategy induces her to form the same posterior beliefs as those she would

have selected with correct priors. To see the intuition behind this result, note that by Bayesian updating, the posterior belief about the good state tomorrow given a signal realization s_j today is given by

$$P_{x_1|s_\lambda}(x_1^2|s_j) = (1-p)\Pr(x_0^1|s_j) + p\Pr(x_0^2|s_j).$$

I show in the proof of Proposition 3.2 that the rationally inattentive investor forms posteriors $\Pr(x_0^i|s_j)$ about x_0^i by scaling the prior about $x_0^i - p_{x_0^i}$ - by some constant $\frac{C_{i,j}}{p_{x_0^i}}$ that is inversely proportional to $p_{x_0^i}$, which induces the latter to vanish. The optimal information strategy thus corrects any prior biases about x_0 and results in a posterior that is objectively correct given the information and Bayesian updating (note that the reasoning does not apply to investors who are biased in their beliefs about tomorrow conditional on a known state today).

In Section 3.1 of Chavchanidze (2021), I prove in a more general setting that as $\lambda \rightarrow 0$, the posterior beliefs converge in probability to the true probability distribution over the states tomorrow when the agent has correct priors (in the setting of this paper, the convergence holds with incorrect priors too). The intuition behind this result is that in the limit, the agent observes correct signals with probability approaching to one and applies a weight of zero to the incorrect probability of the good state tomorrow and a weight of one to the true probability (the latter can easily be seen by inspection, since $e^{-\frac{\beta}{\lambda}KL}$ goes to zero as λ becomes tiny).

When forming posterior beliefs, an increase in the cost of information (equivalently, a decrease in signal informativeness) induces the agent to apply relatively smaller weight to p following a good signal realization and a relatively larger weight following a bad signal realization (compared to the cases with smaller λ). This induces the posterior probability of the good state tomorrow given a good signal today to decrease with increases in λ and the posterior probability of the good state tomorrow given a bad signal today to increase. This property is formalized in the Corollary that follows.

Corollary 3.1. *For small enough λ ,*

$$\begin{aligned} \frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} &< 0 \\ \frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda} &> 0 \end{aligned}$$

and

$$\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} = -\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda}.$$

Proof. See Part B of the Appendix. $\square$

3.3 Asset Pricing with Posterior Beliefs

Let $p_{2|i,\lambda} \equiv P_{x_1|s_\lambda}(x_1^2|s_i)$, $i = 1, 2$. When the representative agent prices the risky and safe assets using posterior beliefs, expressions (3.2) and (3.1) in Section 3.1 become

$$P_{r,s_j} = \beta \left(\left(1 - p_{2|j,\lambda}\right) \frac{1}{g_1} X_1 + p_{2|j,\lambda} \frac{1}{g_2} X_2 \right)$$

and

$$P_{f,s_j} = \beta \left(\left(1 - p_{2|j,\lambda}\right) \frac{1}{g_1} + p_{2|j,\lambda} \frac{1}{g_2} \right),$$

where

$$P_{r,s_j} - P_{f,s_j} = \beta \left(\left(1 - p_{2|j,\lambda}\right) \frac{1}{g_1} (X_1 - 1) + p_{2|j,\lambda} \frac{1}{g_2} (X_2 - 1) \right).$$

Denote $P_{r,s_j} - P_{f,s_j} \equiv \Delta_{s_j}$ and let $\Delta_{x_0^i}$ be the difference between the equilibrium prices of the risky and safe assets in state x_0^i of the complete information economy, $i = 1, 2$, $j = 1, 2$. Then, if we let p_{2,x_0^i} denote the probability of the good state tomorrow in state x_0^i today,

$$\Delta_{x_0^i} = \beta \left(\left(1 - p_{2,x_0^i}\right) \frac{1}{g_1} (X_1 - 1) + p_{2,x_0^i} \frac{1}{g_2} (X_2 - 1) \right)$$

and

$$\begin{aligned} \Delta_{s_j} - \Delta_{x_0^j} &= \beta \left(\left(p_{2|j,\lambda} - p_{2,x_0^j} \right) \frac{1}{g_1} (1 - X_1) + \frac{1}{g_2} (X_2 - 1) \left(p_{2|j,\lambda} - p_{2,x_0^j} \right) \right) = \\ &= \left(p_{2|j,\lambda} - p_{2,x_0^j} \right) \beta \left(\frac{1}{g_1} (1 - X_1) + \frac{1}{g_2} (X_2 - 1) \right). \end{aligned}$$

Since $\beta \left(\frac{1}{g_1} (1 - X_1) + \frac{1}{g_2} (X_2 - 1) \right) > 0$, the sign of $\Delta_{s_j} - \Delta_{x_0^j}$ depends on the sign of $p_{2|j,\lambda} - p_{2,x_0^j}$, where

$$p_{2,x_0^1} = 1 - p$$

$$p_{2,x_0^2} = p$$

by model assumptions and

$$p_{2|1,\lambda} = (1-p) \frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} + p \frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}}$$

$$p_{2|2,\lambda} = (1-p) \frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}} + p \frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}}$$

by Proposition 3.2. This leads to the following proposition:

Proposition 3.3. *Assume λ is small enough so that the representative agent chooses to process information. Let $\Delta_{s_j} = P_{r,s_j} - P_{f,s_j}$ and let $\Delta_{x_0^i}$ be the difference between the equilibrium prices of the risky and safe assets in state x_0^i of the complete information economy, $i = 1, 2$, $j = 1, 2$. Then,*

$$\Delta_{s_j} - \Delta_{x_0^2} < 0$$

$$\Delta_{s_j} - \Delta_{x_0^1} > 0.$$

Proof. $p_{2|j,\lambda} - p_{2,x_0^2} < 0$ and $p_{2|j,\lambda} - p_{2,x_0^1} > 0$, $j = 1, 2$. □

The proposition above says that relative to the complete information economy, the price wedge in the incomplete information economy is smaller in the good time-zero state and larger in the bad time-zero state, which is in line with the findings in Chavchanidze (2021), according to which incompleteness of information induces risk-aversion in the good state and risk-seeking in the bad state relative to the case with complete information. When a bad signal is observed in the good state, the wedge is smaller because the representative agent wrongly believes that the outlook for tomorrow is bad. When a good signal is observed in the good state, the wedge is still smaller relative to the complete information case, because the economy prices in signal noisiness (equivalently, the possibility that the outlook is not as good as suggested by imperfect signals). Alternatively, the price wedge in the perfect information case is higher, because the complete information economy is endowed with fundamental uncertainty only, while the incomplete information economy is endowed with fundamental as well as informational uncertainty.

The analysis for the bad time-zero state mirrors the reasoning above: when a good signal is observed in the bad state, the economy is mistaken into thinking that the outlook for tomorrow is good and the price wedge widens. When a bad signal is observed

in the bad state, the price wedge is still bigger relative to the complete information case, because informational uncertainty prevents economic agents from seeing with certainty that the worst-case scenario is underway, which induces risk-seeking relative to the perfect information case.

The proposition that follows describes how these effects change as the cost of information increases.

Proposition 3.4. *Let P_{r,s_j} and P_{f,s_j} denote, respectively, the prices of the risky and safe assets in the incomplete information economy following a signal realization s_j , $j = 1, 2$. Assume λ is small enough so that the representative agent chooses to process information. Then,*

$$\frac{\partial (P_{r,s_1} - P_{f,s_1})}{\partial \lambda} > 0,$$

$$\frac{\partial (P_{r,s_2} - P_{f,s_2})}{\partial \lambda} < 0$$

and

$$\frac{\partial (P_{r,s_1} - P_{f,s_1})}{\partial \lambda} = - \frac{\partial (P_{r,s_2} - P_{f,s_2})}{\partial \lambda}.$$

Proof. See Part C of the Appendix. □

The proposition above implies that as information becomes more costly to acquire, the effects of incomplete information are amplified when a correct signal is observed: the price wedge corresponding to a bad signal increases (greater risk-seeking relative to the complete information case) and the wedge corresponding to the good signal decreases (greater risk-aversion relative to the complete information case). In contrast, higher λ mitigates deviations from the complete information cases when the signals are incorrect, since the agent does not trust signals as much when λ is higher and gives relatively higher weight to priors.

3.3.1 Remark on the Risk Premium and Low Risk-Free Rate Puzzles

Note that in the incomplete information economy, $\frac{X}{P_{r,s_j}}$ and $\frac{1}{P_{f,s_j}}$ denote, respectively, the return of the risky asset and the risk-free rate when the two assets are priced according to the signal realization s_j . Then, the unconditional risk premium in the incomplete information economy is given by

$$E_{s_\lambda} E_{x_1|s_\lambda} \left[\frac{X}{P_{r,s_\lambda}} - \frac{1}{P_{f,s_\lambda}} \right] = p_{s,\lambda} E_{x_1|s_2} \left[\frac{X}{P_{r,s_2}} - \frac{1}{P_{f,s_2}} \right] + (1 - p_{s,\lambda}) E_{x_1|s_1} \left[\frac{X}{P_{r,s_1}} - \frac{1}{P_{f,s_1}} \right],$$

which is the signal probability-weighted average of the conditional risk premia induced by different signal realizations. When the cost of information changes, so do unconditional signal probabilities, conditional distributions over x_1 and the prices of the two assets. Even though the expression looks complicated, the analytic closed-form tractability should allow us to sharply describe the mechanisms by which different changes induced by λ interact with each other and determine the direction and magnitude of the change in the unconditional risk premium, which could in turn provide insights regarding the risk premium puzzle. Since $\frac{1}{P_{f,s_\lambda}}$ is the risk free rate, analyzing $\frac{1}{P_{f,s_\lambda}}$ alone in a similar way might illustrate some mechanisms relevant to the low risk-free rate puzzle.

4 Conclusion

I build on my previous work in Chavchanidze (2021), in which I showed that under fixed asset prices, a rationally inattentive investor with uniform priors is risk-averse in the good time-zero state and risk-seeking in the bad time-zero state relative to the case with complete information. I extend the result by showing that the same holds for non-uniform priors and flexible prices. Results in this paper are significant beyond their immediate setting, since I obtain unconditional signal probabilities and posterior beliefs in closed form and for general two-state distribution, which could help facilitate analysis of informational frictions in more general cases, such as in multi-period models or binomial asset pricing setting.

5 References

- Batchuluun, Altantsetseg, Yulei Luo, and Eric R Young (2019). “Portfolio Choice with Information-Processing Limits”. In: *Annals of Economics and Finance* 20 (1), pp. 137–162.
- Chavchanidze, Giorgi (2021). “Cognitive Limitations and Behavioral Biases in the Asset Pricing Context”. MA thesis. CERGE-EI.
- Luo, Yulei (2006). “Rational Inattention, Portfolio Choice, and the Equity Premium”. Job Market Paper.
- Mackowiak, Bartosz, Filip Matejka, and Mirko Wiederholt (2020). “Rational Inattention: A Review”. CEPR Discussion Paper 15408.
- Matejka, Filip and Alisdair McKay (2015). “Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model”. In: *American Economic Review* 105 (1), pp. 272–298.
- Shefrin, Hersh (2008). *Behavioral Approach To Asset Pricing*. Academic Press.

6 Appendix

The appendix consists of the proofs of the propositions discussed in the body of the paper. Any result that is classified as a *Lemma* in the appendix is an intermediate result needed for proving one of the propositions that appear in the body.

Part A

Lemma 6.1. *Let $p_{s,\lambda}$ denote the probability of observing a good signal under the optimal information strategy and the cost of information λ . Assume λ is small enough so that $0 < p_{s,\lambda} < 1$. Then,*

$$p_{s,\lambda} = \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL}}{1 - e^{-\frac{\beta}{\lambda}KL}}.$$

Proof. Let p_s denote the probability of observing a good signal as a choice variable under the control of the optimizing agent. By Corrolary 2 in Matejka and McKay (2015), under the assumption that $0 < p_{s,\lambda} < 1$ (which holds for small enough cost of information λ), the optimal choice of p_s satisfies

$$(1 - p_0) \frac{e^{-\frac{\beta}{\lambda}KL}}{1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}} + p_0 \frac{1}{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL}} = 1 \quad (6.1)$$

and

$$(1 - p_0) \frac{1}{1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}} + p_0 \frac{e^{-\frac{\beta}{\lambda}KL}}{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL}} = 1. \quad (6.2)$$

Multiplying both sides of (6.1) and (6.2) by $p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL}$ and doing trivial algebraic rearrangments, we get, respectively,

$$\frac{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL}}{1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}} = \frac{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL} - p_0}{(1 - p_0) e^{-\frac{\beta}{\lambda}KL}} \quad (6.3)$$

and

$$\frac{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL}}{1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}} = \frac{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL} - p_0 e^{-\frac{\beta}{\lambda}KL}}{(1 - p_0)}.$$

Note that the LHS in both expressions above are the same and so

$$\frac{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL} - p_0}{(1 - p_0) e^{-\frac{\beta}{\lambda}KL}} = \frac{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda}KL} - p_0 e^{-\frac{\beta}{\lambda}KL}}{(1 - p_0)},$$

which implies by straightforward algebra that

$$\begin{aligned}
p_s + (1 - p_s)e^{-\frac{\beta}{\lambda}KL} - p_0 &= e^{-\frac{\beta}{\lambda}KL}p_s + (1 - p_s)e^{-2\frac{\beta}{\lambda}KL} - p_0e^{-2\frac{\beta}{\lambda}KL} \\
p_s \left(1 - 2e^{-\frac{\beta}{\lambda}KL} + e^{-2\frac{\beta}{\lambda}KL}\right) &= -e^{-\frac{\beta}{\lambda}KL} + p_0 + e^{-2\frac{\beta}{\lambda}KL} - p_0e^{-2\frac{\beta}{\lambda}KL} \\
p_s \left(1 - e^{-\frac{\beta}{\lambda}KL}\right)^2 &= p_0 \left(1 - e^{-2\frac{\beta}{\lambda}KL}\right) + e^{-\frac{\beta}{\lambda}KL} \left(e^{-\frac{\beta}{\lambda}KL} - 1\right),
\end{aligned}$$

where the RHS on the third line satisfies

$$\begin{aligned}
RHS &= p_0 \left(1 - e^{-\frac{\beta}{\lambda}KL}\right) \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL} \left(1 - e^{-\frac{\beta}{\lambda}KL}\right) = \\
&= \left(1 - e^{-\frac{\beta}{\lambda}KL}\right) \left(p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL}\right).
\end{aligned}$$

Hence,

$$p_s \left(1 - e^{-\frac{\beta}{\lambda}KL}\right)^2 = \left(1 - e^{-\frac{\beta}{\lambda}KL}\right) \left(p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL}\right)$$

and

$$p_s = \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL}}{1 - e^{-\frac{\beta}{\lambda}KL}}.$$

□

Corollary 6.1. $\exists \bar{\lambda} > 0$ such that the agent does not process any information for $\lambda \geq \bar{\lambda}$.

Proof. Note first that since $\lim_{\lambda \rightarrow \infty} e^{-\frac{\beta}{\lambda}KL} = 1$ and $p_0 > \frac{1}{2}$,

$$\begin{aligned}
\lim_{\lambda \rightarrow \infty} \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL}}{1 - e^{-\frac{\beta}{\lambda}KL}} &= \frac{p_0 \left(1 + \lim_{\lambda \rightarrow \infty} e^{-\frac{\beta}{\lambda}KL}\right) - \lim_{\lambda \rightarrow \infty} e^{-\frac{\beta}{\lambda}KL}}{1 - \lim_{\lambda \rightarrow \infty} e^{-\frac{\beta}{\lambda}KL}} = \\
&= \frac{2p_0 - 1}{1 - \lim_{\lambda \rightarrow \infty} e^{-\frac{\beta}{\lambda}KL}} = \infty
\end{aligned} \tag{6.4}$$

and since $\lim_{\lambda \rightarrow 0} e^{-\frac{\beta}{\lambda}KL} = 0$,

$$\begin{aligned}
\lim_{\lambda \rightarrow 0} \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) - e^{-\frac{\beta}{\lambda}KL}}{1 - e^{-\frac{\beta}{\lambda}KL}} &= \frac{p_0 \left(1 + \lim_{\lambda \rightarrow 0} e^{-\frac{\beta}{\lambda}KL}\right) - \lim_{\lambda \rightarrow 0} e^{-\frac{\beta}{\lambda}KL}}{1 - \lim_{\lambda \rightarrow 0} e^{-\frac{\beta}{\lambda}KL}} = \\
&= p_0 < 1.
\end{aligned} \tag{6.5}$$

Next, set $\frac{\partial \left(e^{-\frac{\beta}{\lambda} KL} \right)}{\partial \lambda} \equiv e'$ and observe that since $e' = \beta KL \frac{e^{-\frac{\beta}{\lambda} KL}}{\lambda^2} > 0$,

$$\begin{aligned} & \frac{\partial}{\partial \lambda} \left(\frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}} \right) = \\ &= \frac{e' (p_0 - 1) \left(1 - e^{-\frac{\beta}{\lambda} KL} \right) + \left(p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL} \right) e'}{\left(1 - e^{-\frac{\beta}{\lambda} KL} \right)^2} = \\ &= \frac{e' (2p_0 - 1)}{\left(1 - e^{-\frac{\beta}{\lambda} KL} \right)^2} > 0. \end{aligned} \tag{6.6}$$

Suppose now that the agent chooses to process information for all $\lambda > 0$. Then, $0 < p_{s,\lambda} < 1$ (otherwise, the agent observes only one signal realization with probability 1, in which case information strategy is completely uninformative and it is not optimal to incur costs for signals that do not inform about the state of the world). By Lemma 6.1, $0 < p_{s,\lambda} < 1$ implies $p_{s,\lambda} = \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}}$, but since $\frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}}$ is a continuous function of λ , (6.5), (6.4) and (6.6) together imply that $\exists \bar{\lambda} > 0$ (call it $\bar{\lambda}$) such that

$$\frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}} = 1,$$

which is a contradiction. Hence, the agent does not process any information whenever the cost of information parameter equals $\bar{\lambda}$ and (6.6) implies the same holds $\forall \lambda > \bar{\lambda}$. $\square$

Proposition 3.1. *Let $p_{s,\lambda}$ denote the probability of observing a good signal under the optimal information strategy and the cost of information λ and let $p_0 > \frac{1}{2}$ be the prior probability of the good state at time 0. Then,*

i) $\exists \bar{\lambda} > 0$ such that the agent does not process any information for $\lambda \geq \bar{\lambda}$.

ii) $\lim_{\lambda \rightarrow 0} p_{s,\lambda} = p_0$,

iii) For small enough λ , $p_{s,\lambda} = \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}}$,

iv) For small enough λ , $\frac{\partial p_{s,\lambda}}{\partial \lambda} > 0$.

Proof. iii) is proved as Lemma 6.1 and i) is proved as Corollary 6.1. See (6.5) for ii)

and (6.6) for iv). □

Part B

Lemma 6.2. *Let $p_{s,\lambda}$ denote the probability of observing a good signal under the optimal information strategy and the cost of information λ . Assume $p_0 > \frac{1}{2}$ and λ is small enough so that $0 < p_{s,\lambda} < 1$. Then,*

$$(1 - p_0) \left(\frac{p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}{1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL}} \right)^2 > p_0.$$

Proof. By Lemma 6.1,

$$p_{s,\lambda} = \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}}$$

and

$$1 - p_{s,\lambda} = \frac{1 - p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right)}{1 - e^{-\frac{\beta}{\lambda} KL}},$$

which implies that

$$\begin{aligned} & p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} = \\ &= \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL} + e^{-\frac{\beta}{\lambda} KL} - e^{-\frac{\beta}{\lambda} KL} p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right)}{1 - e^{-\frac{\beta}{\lambda} KL}} = \\ &= \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) \left(1 - e^{-\frac{\beta}{\lambda} KL} \right)}{1 - e^{-\frac{\beta}{\lambda} KL}} = p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right). \end{aligned} \tag{6.7}$$

Hence, by (6.3),

$$\begin{aligned} & \frac{p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}{1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL}} = \frac{p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} - p_0}{(1 - p_0) e^{-\frac{\beta}{\lambda} KL}} = \\ &= \frac{p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - p_0}{(1 - p_0) e^{-\frac{\beta}{\lambda} KL}} = \frac{p_0}{1 - p_0}, \end{aligned}$$

which implies that for $p_0 > 1/2$

$$(1-p_0) \left(\frac{p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}{1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL}} \right)^2 = p_0 \frac{p_0}{1-p_0} > p_0.$$

□

Proposition 3.2. *Let s_j denote a signal realization that suggests the current state is j , $j = 1, 2$. Then,*

$$P_{x_1|s_\lambda}(x_1^2|s_2) = (1-p) \frac{e^{-\frac{\beta}{\lambda} KL}}{1 + e^{-\frac{\beta}{\lambda} KL}} + p \frac{1}{1 + e^{-\frac{\beta}{\lambda} KL}}$$

and

$$P_{x_1|s_\lambda}(x_1^2|s_1) = (1-p) \frac{1}{1 + e^{-\frac{\beta}{\lambda} KL}} + p \frac{e^{-\frac{\beta}{\lambda} KL}}{1 + e^{-\frac{\beta}{\lambda} KL}}.$$

Proof. By Proposition 3.1 and Sections 3.2 and 4.2 in Chavchanidze (2021), the posterior beliefs with log-utility and 2 states are given by

$$P_{x_1|s_\lambda}(x_1^2|s_2) = (1-p)(1-p_0) \frac{e^{-\frac{\beta}{\lambda} KL}}{1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL}} + pp_0 \frac{1}{p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}$$

and

$$P_{x_1|s_\lambda}(x_1^2|s_1) = (1-p)(1-p_0) \frac{1}{1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL}} + pp_0 \frac{e^{-\frac{\beta}{\lambda} KL}}{p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}.$$

By (6.7),

$$p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} = p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) \quad (6.8)$$

and by Lemma 6.1,

$$\begin{aligned} & 1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} = \\ &= \frac{1 - p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) + e^{-\frac{\beta}{\lambda} KL} p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) - e^{-\frac{\beta}{\lambda} KL} e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}} = \\ &= \frac{-p_0 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) \left(1 - e^{-\frac{\beta}{\lambda} KL} \right) + \left(1 + e^{-\frac{\beta}{\lambda} KL} \right) \left(1 - e^{-\frac{\beta}{\lambda} KL} \right)}{1 - e^{-\frac{\beta}{\lambda} KL}} = \end{aligned}$$

$$= -p_0 \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) + \left(1 + e^{-\frac{\beta}{\lambda}KL}\right) = (1 - p_0) \left(1 + e^{-\frac{\beta}{\lambda}KL}\right). \quad (6.9)$$

Inserting (6.8) and (6.9) back into the posterior beliefs yields the claim of the proposition. $\square$

Corollary 3.1. *For small enough λ ,*

$$\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} < 0$$

$$\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda} > 0$$

and

$$\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} = -\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda}.$$

Proof. Note that

$$\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} + \frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}} = 1 \quad (6.10)$$

and so

$$\frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) + \frac{\partial}{\partial \lambda} \left(\frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) = 0. \quad (6.11)$$

Hence,

$$\begin{aligned} \frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} &= (1-p) \frac{\partial}{\partial \lambda} \left(\frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) + p \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) = \\ &= -(1-p) \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) + p \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) = \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) (2p-1) \end{aligned}$$

and

$$\begin{aligned} \frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda} &= (1-p) \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) + p \frac{\partial}{\partial \lambda} \left(\frac{e^{-\frac{\beta}{\lambda}KL}}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) = \\ &= (1-p) \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) - p \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) = \frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda}KL}} \right) (1-2p), \end{aligned}$$

where the second equality in both equations follows by (6.11). This shows

$$\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} = -\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda}.$$

Lastly, note that $p > \frac{1}{2}$ by assumption and

$$\frac{\partial}{\partial \lambda} \left(\frac{1}{1 + e^{-\frac{\beta}{\lambda} KL}} \right) = - \frac{\beta K L e^{-\frac{\beta}{\lambda} KL}}{\lambda^2 \left(1 + e^{-\frac{\beta}{\lambda} KL} \right)^2} < 0,$$

which together imply that $\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} < 0$ and $\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_1)}{\partial \lambda} > 0$. $\square$

Part C

Proposition 3.4. *Let P_{r,s_j} and P_{f,s_j} denote, respectively, the prices of the risky and safe assets in the incomplete information economy following a signal realization s_j , $j = 1, 2$. Assume λ is small enough so that the representative agent chooses to process information. Then,*

$$\frac{\partial (P_{r,s_1} - P_{f,s_1})}{\partial \lambda} > 0,$$

$$\frac{\partial (P_{r,s_2} - P_{f,s_2})}{\partial \lambda} < 0$$

and

$$\frac{\partial (P_{r,s_1} - P_{f,s_1})}{\partial \lambda} = - \frac{\partial (P_{r,s_2} - P_{f,s_2})}{\partial \lambda}.$$

Proof. Let $p_{2|j,\lambda} \equiv P_{x_1|s_\lambda}(x_1^2|s_j)$ and recall that

$$P_{r,s_j} - P_{f,s_j} = \beta \left((1 - p_{2|j,\lambda}) \frac{1}{g_1} (X_1 - 1) + p_{2|j,\lambda} \frac{1}{g_2} (X_2 - 1) \right),$$

which implies

$$\begin{aligned} \frac{\partial (P_{r,s_j} - P_{f,s_j})}{\partial \lambda} &= \frac{\partial (P_{r,s_j} - P_{f,s_j})}{\partial p_{2|j,\lambda}} \frac{\partial p_{2|j,\lambda}}{\partial \lambda} = \\ &= \beta \left(-\frac{1}{g_1} (X_1 - 1) + \frac{1}{g_2} (X_2 - 1) \right) \frac{\partial p_{2|j,\lambda}}{\partial \lambda}. \end{aligned}$$

By model assumptions, $\beta \left(-\frac{1}{g_1} (X_1 - 1) + \frac{1}{g_2} (X_2 - 1) \right) > 0$ and by Corollary 3.1,

$$\frac{\partial p_{2|2,\lambda}}{\partial \lambda} < 0, \quad \frac{\partial p_{2|1,\lambda}}{\partial \lambda} > 0$$

and

$$\frac{\partial p_{2|2,\lambda}}{\partial \lambda} = -\frac{\partial p_{2|1,\lambda}}{\partial \lambda}.$$

The claim of the Proposition is then immediate. $\square$

Part D

In this last part of the Appendix, I show how Corollary 3.1 can be proved without knowing an explicit expression for $p_{s,\lambda}$.¹

Lemma 6.3. *Let $p_{s,\lambda}$ denote the probability of observing a good signal under the optimal information strategy and the cost of information λ . Then,*

$$\begin{aligned} \frac{\partial p_{s,\lambda}}{\partial \lambda} = & -\frac{1}{\lambda^2} \frac{\beta KL (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}} \left(-(1 - p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} + p_0 \right) \times \\ & \times \left((1 - p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} e^{-\frac{\beta}{\lambda} KL} + p_0 \right)^{-1}. \end{aligned}$$

Proof. Let p_s denote the probability of observing a good signal as a choice variable under the control of the optimizing agent. By Corollary 2 in Matejka and McKay (2015), the optimal choice of p_s under small enough cost of information λ satisfies

$$(1 - p_0) \frac{e^{-\frac{\beta}{\lambda} KL}}{1 - p_s + p_s e^{-\frac{\beta}{\lambda} KL}} + p_0 \frac{1}{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda} KL}} = 1, \quad (6.12)$$

i.e. (6.12) is the first-order condition of the optimal signal selection problem. Taking the total differential of the expression above w.r.t λ and p_s , we get

$$\frac{\partial LHS}{\partial p_s} dp_s + \frac{\partial LHS}{\partial \lambda} d\lambda = 0, \quad (6.13)$$

¹Even though $p_{s,\lambda}$ can be obtained in closed form in the setting of this paper, in general this is not always feasible. The proofs that follow illustrate a general approach that can be useful for determining the signs of the derivatives of posterior probabilities w.r.t λ in more general cases.

where LHS denotes the left-hand side of (6.12) and $\frac{\partial LHS}{\partial p_s}$ and $\frac{\partial LHS}{\partial \lambda}$ are given by

$$\begin{aligned}
\frac{\partial LHS}{\partial p_s} &= (1-p_0) \frac{e^{-\frac{\beta}{\lambda}KL} \left(1 - e^{-\frac{\beta}{\lambda}KL}\right)}{\left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right)^2} + p_0 \frac{1 - e^{-\frac{\beta}{\lambda}KL}}{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} = \\
&= (1-p_0) \frac{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right)^2} \frac{e^{-\frac{\beta}{\lambda}KL} \left(1 - e^{-\frac{\beta}{\lambda}KL}\right)}{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} + p_0 \frac{1 - e^{-\frac{\beta}{\lambda}KL}}{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} = \\
&= \frac{1 - e^{-\frac{\beta}{\lambda}KL}}{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} \left((1-p_0) \frac{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + p_0 \right) \quad (6.14)
\end{aligned}$$

and

$$\begin{aligned}
\frac{\partial LHS}{\partial \lambda} &= (1-p_0) \frac{\beta KL \frac{1}{\lambda^2} e^{-\frac{\beta}{\lambda}KL} \left(p_s e^{-\frac{\beta}{\lambda}KL}\right) - \left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right) \beta KL \frac{1}{\lambda^2} e^{-\frac{\beta}{\lambda}KL}}{\left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right)^2} + \\
&\quad + p_0 \frac{\beta KL (1-p_s) e^{-\frac{\beta}{\lambda}KL}}{\lambda^2 \left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} = \\
&= -(1-p_0) \frac{1}{\lambda^2} \frac{\beta KL (1-p_s) e^{-\frac{\beta}{\lambda}KL}}{\left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right)^2} + p_0 \frac{1}{\lambda^2} \frac{\beta KL (1-p_s) e^{-\frac{\beta}{\lambda}KL}}{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} = \\
&= \frac{1}{\lambda^2} \frac{\beta KL (1-p_s) e^{-\frac{\beta}{\lambda}KL}}{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2} \left(-(1-p_0) \frac{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_s + p_s e^{-\frac{\beta}{\lambda}KL}\right)^2} + p_0 \right). \quad (6.15)
\end{aligned}$$

(6.13) gives an implicit expression for the required change in the choice variable p_s (in response to an exogenous change in λ) so that the information strategy remains optimal. Hence, putting together (6.13), (6.14) and (6.15), we get

$$\frac{\partial p_{s,\lambda}}{\partial \lambda} = \frac{dp_s}{d\lambda} = - \frac{\frac{\partial LHS}{\partial \lambda}}{\frac{\partial LHS}{\partial p_s}} =$$

$$\begin{aligned}
&= -\frac{1}{\lambda^2} \frac{\beta KL (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}{1 - e^{-\frac{\beta}{\lambda} KL}} \left(- (1 - p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} + p_0 \right) \times \\
&\quad \times \left((1 - p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} e^{-\frac{\beta}{\lambda} KL} + p_0 \right)^{-1}.
\end{aligned}$$

□

Lemma 6.4. *Let $P_{x_1|s_\lambda}(x_1^2|s_2)$ denote the probability of the good state tomorrow conditional on observing a good signal today. Then,*

$$\frac{\partial P_{x_1|s_\lambda}(x_1^2|s_2)}{\partial \lambda} < 0.$$

Proof. Consider $P_{x_1|s_\lambda}(x_1^2|s_2)$ as a function of λ and $p_{s,\lambda}$, i.e. for the purposes of the proof, set $P_{x_1|s_\lambda}(x_1^2|s_2) \equiv P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})$ and let $\frac{\tilde{\partial} P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\tilde{\partial} \lambda}$ denote a partial derivative of $P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})$ w.r.t to its first argument (equivalently, a partial derivative of $P_{x_1|s_\lambda}(x_1^2|s_2)$ w.r.t. λ conditional on keeping $p_{s,\lambda}$ fixed). Then, by the chain rule of differentiation,

$$\frac{\partial P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\partial \lambda} = \frac{\tilde{\partial} P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\tilde{\partial} \lambda} + \frac{\partial P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\partial p_{s,\lambda}} \frac{\partial p_{s,\lambda}}{\partial \lambda}. \quad (6.16)$$

Note that the LHS of (6.12) given by

$$(1 - p_0) \frac{e^{-\frac{\beta}{\lambda} KL}}{1 - p_s + p_s e^{-\frac{\beta}{\lambda} KL}} + p_0 \frac{1}{p_s + (1 - p_s) e^{-\frac{\beta}{\lambda} KL}} \quad (6.17)$$

is analogous to

$$P_{x_1|s_\lambda}(\lambda, p_{s,\lambda}) = (1 - p) (1 - p_0) \frac{e^{-\frac{\beta}{\lambda} KL}}{1 - p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL}} + p p_0 \frac{1}{p_{s,\lambda} + (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL}}$$

in that we get $P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})$ by multiplying the first summand in (6.17) by $(1 - p)$ and

the second summand - by p . Hence, by analogy to (6.14) and (6.15), we get, respectively,

$$\begin{aligned} \frac{\partial P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\partial p_{s,\lambda}} &= \frac{1 - e^{-\frac{\beta}{\lambda}KL}}{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2} \times \\ &\times \left((1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + pp_0 \right) \end{aligned}$$

and

$$\begin{aligned} \frac{\tilde{\partial} P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\tilde{\partial} \lambda} &= \frac{1}{\lambda^2} \frac{\beta KL (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda}KL}}{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2} \times \\ &\times \left(-(1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} + pp_0 \right). \end{aligned} \quad (6.18)$$

Hence, by Lemma 6.3,

$$\begin{aligned} \frac{\partial P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\partial p_{s,\lambda}} \frac{\partial p_{s,\lambda}}{\partial \lambda} &= - \frac{(1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + pp_0}{(1-p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + p_0} \times \\ &\times \frac{1}{\lambda^2} \frac{\beta KL (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda}KL}}{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2} \left(-(1-p_0) \frac{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1 - p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} + p_0 \right), \end{aligned}$$

which in turn implies by (6.16) and (6.18) that

$$\frac{\partial P_{x_1|s_\lambda}(\lambda, p_{s,\lambda})}{\partial \lambda} = \frac{1}{\lambda^2} \frac{\beta KL (1 - p_{s,\lambda}) e^{-\frac{\beta}{\lambda}KL}}{\left(p_{s,\lambda} + (1 - p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2} \times$$

$$\begin{aligned}
& \times \left(- (1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} + pp_0 - \right. \\
& \quad \frac{(1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} e^{-\frac{\beta}{\lambda} KL} + pp_0}{(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} e^{-\frac{\beta}{\lambda} KL} + p_0} \times \\
& \quad \left. \times \left(- (1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} + p_0 \right) \right) \equiv \\
& \quad \equiv A \times (B_1 - B_2 B_3).
\end{aligned}$$

It is obvious that $A > 0$. Hence, if we can show that $(B_1 - B_2 B_3) < 0$, the proof will be complete. This will be accomplished next. Note that since $\frac{1}{2} < p < 1$, $pp_0 > (1-p)p_0$ and so

$$\begin{aligned}
B_1 &= - (1-p)(1-p_0) \frac{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_s + p_s e^{-\frac{\beta}{\lambda} KL} \right)^2} + pp_0 < \\
&< - (1-p)(1-p_0) \frac{\left(p_s + (1-p_s) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_s + p_s e^{-\frac{\beta}{\lambda} KL} \right)^2} + (1-p)p_0 = \\
&= (1-p) \left(- (1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda}) e^{-\frac{\beta}{\lambda} KL} \right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda} e^{-\frac{\beta}{\lambda} KL} \right)^2} + p_0 \right) = (1-p) B_3
\end{aligned}$$

and

$$\begin{aligned}
B_2 &= \frac{(1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + pp_0}{(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + p_0} > \\
&> \frac{(1-p)(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + (1-p)p_0}{(1-p_0) \frac{\left(p_{s,\lambda} + (1-p_{s,\lambda})e^{-\frac{\beta}{\lambda}KL}\right)^2}{\left(1-p_{s,\lambda} + p_{s,\lambda}e^{-\frac{\beta}{\lambda}KL}\right)^2} e^{-\frac{\beta}{\lambda}KL} + p_0} = (1-p).
\end{aligned}$$

Combining these with Lemma 6.2, we get that

$$B_1 - B_2 B_3 < (1-p) B_3 - (1-p) B_3 = 0,$$

as desired. □